

XIAOCHUN NIU

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ACADEMIC POSITIONS

Postdoctoral Researcher	<i>July 2024–Present</i>
The Fuqua School of Business, Duke University	Durham, NC
Hosted by Prof. Jiaming Xu	

EDUCATION

Ph.D. in Industrial Engineering and Management Sciences	<i>Sept. 2019–June 2024</i>
Northwestern University	Evanston, IL
Advised by Prof. Ermin Wei and Prof. Julia Gaudio	
B.S. in Mathematics	<i>Sept. 2015–June 2019</i>
Nanjing University	Nanjing, China
National Elite Program (Honors Program)	

RESEARCH INTERESTS

Operations research, applied probability, high-dimensional statistics, optimization, machine learning, artificial intelligence, data analytics, and network science.

PUBLICATIONS AND PREPRINTS

Journal

- J1. Xiaochun Niu, Tselil Schramm, and Jiaming Xu. “High-dimensional Procrustes Matching via Tree Counts.” 2026+
- J2. Weijia Li, Xiaochun Niu, Yehua Wei, and Jiaming Xu. “[Optimality of Random Regular Graphs for Sparse Network Designs](#).” *Submitted*. 2026+
 - Accepted for presentation at 2026 INFORMS Manufacturing and Service Operations Management (MSOM) Annual Conference
- J3. Xiaochun Niu, Lili Su, Jiaming Xu, and Pengkun Yang. “[Learning with Shared Representations: Statistical Rates and Efficient Algorithms](#).” *Submitted*. 2026+
 - Oral presentation at International Workshop on Federated Foundation Models in conjunction with NeurIPS 2024 (FL@FM-NeurIPS’24)
- J4. Julia Gaudio, Charlie Guan, Xiaochun Niu, and Ermin Wei. “[Exact Label Recovery in Euclidean Random Graphs](#).” *Annals of Applied Probability*, 2026.
- J5. Xiaochun Niu and Ermin Wei. “[DISH: A Distributed Hybrid Optimization Method Leveraging System Heterogeneity](#).” *IEEE Transactions on Signal Processing*, 2024.
 - Extended abstract at Annual Allerton Conference on Communication, Control, and Computing (Allerton), 2022.
- J6. Xiaochun Niu and Ermin Wei. “[FedHybrid: A Hybrid Federated Optimization Method for Heterogeneous Clients](#).” *IEEE Transactions on Signal Processing*, 2023.

Conference

- C1. Weijia Li, Xiaochun Niu, Yehua Wei, and Jiaming Xu. “[Optimality of Random Regular Graphs for Sparse Network Designs](#).” *ACM Conference on Economics and Computation (EC)*, 2026.
- C2. Lihui Yi, Xiaochun Niu, and Ermin Wei. “[Incentive Analysis for Agent Participation in Federated Learning](#).” *IEEE Conference on Decision and Control (CDC)*, 2025.
- C3. Julia Gaudio, Xiaochun Niu, and Ermin Wei. “[Exact Community Recovery in the Geometric Stochastic Block Model](#).” *ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2024.
- **Nemhauser Prize for Best Student Paper**, 2024
- C4. Zhenyu Sun, Xiaochun Niu, and Ermin Wei. “[Understanding Generalization of Federated Learning via Stability: Heterogeneity Matters](#).” *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024
- C5. Xiaochun Niu and Ermin Wei. “[DISH: A Distributed Hybrid Primal-Dual Optimization Framework to Utilize System Heterogeneity](#).” *IEEE Conference on Decision and Control (CDC)*, 2022.

SELECTED HONORS & AWARDS

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| • Rising Stars in Computational and Data Sciences, UT Austin | 2024 |
| • Nemhauser Prize for Best Student Paper, Northwestern University | 2024 |
| • SIAM Student Travel Award, SODA24 | 2024 |
| • Terminal Year Fellowship, Northwestern University | 2023 |
| • Arthur P. Hurter Award for Academic Excellence, Northwestern University | 2020 |
| • Walter P. Murphy Graduate Fellowship, Northwestern University | 2019 |
| • Excellent Graduate Award, Nanjing University | 2019 |
| • UCLA CSST Research Program Scholarship, University of California, Los Angeles | 2018 |
| • National Scholarship (<i>Top 2%</i>), Nanjing University | 2016 |

INVITED TALKS

Optimality of Random Regular Graphs for Sparse Network Designs

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| • INFORMS MSOM Conference, Boston, MA | July 2026 |
| • INFORMS Annual Meeting, Atlanta, GA | Oct. 2025 |

Learning with Shared Representations: Statistical Rates and Efficient Algorithms

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| • INFORMS Applied Probability Society Conference, Atlanta, GA | June 2025 |
| • International Workshop on Federated Foundation Models (NeurIPS 2024), Vancouver | Dec. 2024 |
| • INFORMS Annual Meeting, Seattle, WA | Oct. 2024 |
| • Annual Allerton Conference on Communication, Control, and Computing (Allerton), UIUC | Sept. 2024 |

Exact Community Recovery in the Geometric Stochastic Block Model

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| • IEMS-SIAM Student Seminar, Northwestern University | May 2024 |
| • Rising Stars in Computational and Data Sciences, UT Austin | April 2024 |

- Institute for Data, Econometrics, Algorithms, and Learning (IDEAL) Special Program on Network and Inference, Toyota Technological Institute at Chicago (TTIC) *Jan. 2024*
- ACM-SIAM Symposium on Discrete Algorithms (SODA), Alexandria, VA *Jan. 2024*

DISH: A Distributed Hybrid Optimization Method Leveraging System Heterogeneity

- INFORMS Annual Meeting, Phoenix, AZ *Oct. 2023*
- INFORMS Annual Meeting, Indianapolis, IN *Oct. 2022*
- INFORMS Optimization Society Conference, Greenville, SC *Mar. 2022*

TEACHING EXPERIENCE

Duke University

- EGRMGMT 580, Decision Models, Instructor *Spring 2026*

Northwestern University

- IEMS 435, Stochastic Simulation, Grader *Spring 2022*
- IEMS 344, Whole Brain Leadership, Teaching Assistant *Fall 2021*

INDUSTRIAL EXPERIENCE

Meta Platforms

Menlo Park, CA

Software Engineer Intern, Social Impact Machine Learning Team

June 2022–Sept. 2022

- Improved Facebook's donation-specific newsfeed ranking algorithm by implementing new features and optimizing neural network architecture.

The Home Depot

Atlanta, GA (remote)

Collaborative Data Project

Sept. 2022–Mar. 2023

- Evaluated multi-armed bandit methods on dynamic ad placement datasets with frequent arm adjustments.

PROFESSIONAL SERVICE AND MENTORING

- Reviewer:
 - Journals: Operations Research, Annals of Applied Probability, Journal of Machine Learning Research, Journal of Computational and Graphical Statistics, IEEE Transactions on Information Theory, IEEE Transactions on Automatic Control, IEEE Transactions on Signal Processing, IEEE Transactions on Signal and Information Processing over Networks
 - Conferences: IEEE Conference on Decision and Control, Midwest Machine Learning Symposium, American Control Conference
- Session Organizer: IEEE CDC 2022 (Distributed Optimization and Learning for Networked Systems)
- Mentor: NCSSM Mentorship Program 2026, IDEAL Get Ready for Research Workshop 2023, INFORMS Mentorship Program, Northwestern English Language Program